

Unsupervised Learning: Formal Definition & Other Types

Formal Definition

- **Supervised Learning:** The data provided is a set of **(x, y)** pairs, where **x** is the input and **y** is the correct output label.
 - **Unsupervised Learning:** The data provided is only a set of inputs **(x)**, with **no corresponding output labels**. The goal is for the algorithm to discover interesting structures or patterns within this unlabeled data.
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Types of Unsupervised Learning

Beyond clustering, there are other important types of unsupervised learning algorithms.

1. Clustering (Recap):

- Groups similar data points together.
- *Example:* Grouping news articles or segmenting customers.

2. Anomaly Detection:

- Used to identify rare or unusual data points/events.
- *Example: Fraud detection.* An algorithm can analyze many financial transactions to learn what a "normal" transaction looks like and then flag unusual ones that might be fraudulent.

3. Dimensionality Reduction:

- Compresses a large dataset into a much smaller one while losing as little information as possible.
 - This is useful for data visualization, saving storage space, and speeding up other learning algorithms.
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Reinforcing the Concepts (Quiz Summary)

Here's how to categorize different problems:

- **Spam Filtering (Supervised):** Requires a dataset of emails already labeled as "spam" or "not spam."
- **Grouping News Articles (Unsupervised):** Uses a clustering algorithm to find related articles without any pre-existing labels.
- **Market Segmentation (Unsupervised):** Uses a clustering algorithm to discover customer groups automatically from unlabeled data.
- **Diabetes Diagnosis (Supervised):** Requires a dataset of patient records labeled as

"diabetes" or "no diabetes," similar to the breast cancer classification example.